

Lecture 10.1

Chapter 4 Lagrange's Theorem

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In this lecture, we will introduce some new object called cosets, which are needed to prove Lagrange's Theorem. We will see some examples and study their properties.

The GOAL is to prove Lagrange's Thm:

THEOREM 4.4. LAGRANGE'S THEOREM. *Let H be a subgroup of a finite group G . Then the order of G is a multiple of the order of H .*

What are cosets?



Definition 4.1. Let H be a subgroup of a group G and $a \in G$. The set $Ha = \{ha \mid h \in H\}$ is called the **right coset** of H in G **generated** by a , while $aH = \{ah \mid h \in H\}$ is called the **left coset** of H in G **generated** by a .

Remarks

Observe that in **additive notation** the left and right cosets are:

$$H \pm a = \{h \pm a \mid h \in H\}, \quad a \pm H = \{a \pm h \mid h \in H\}$$

It is important to notice that **cosets in a group G is a subset of G .**

Cosets are not elements in G !

Example 4.2. Find all the left and right cosets of $H = \{\varepsilon, (1\ 3)\}$ in S_3 .

Recall that $S_3 = \{\varepsilon, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Then we have

Take $a = (1\ 2)$.

$$aH = (1\ 2)H = \{(1\ 2), (1\ 2)(1\ 3)\} \stackrel{\oplus}{=} \{(1\ 2), (1\ 3\ 2)\}$$

$$\stackrel{\oplus}{(1\ 2)(1\ 3)} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = (1\ 3\ 2)$$

$$Ha = H(1\ 2) = \{(1\ 2), (1\ 3)(1\ 2)\} = \{(1\ 2), (1\ 2\ 3)\}$$

$$(1\ 3)(1\ 2) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (1\ 2\ 3)$$

Exercises. Compute $H(2\ 3)$ and

$$(2\ 3)H.$$

Example 4.3. Find all the cosets of the subgroup $4\mathbb{Z}$ in the additive group $\mathbb{Z}$. In this case, since $\mathbb{Z}$ is an abelian group, it is enough to compute the right (or the left) cosets.

For $a \in \mathbb{Z}$, we want to compute the left (or right) generated by a :

$$4\mathbb{Z} + a = a + 4\mathbb{Z}$$

$n := 4q + a$, $q \in \mathbb{Z}$ $\longrightarrow$ a is the remainder when n is divided by 4.

Notice that a is the remainder when n is divided by 4. Recall that there are only 4 possible remainders when divided by 4: 0, 1, 2, 3.

Thus, we only have 4 cosets:

$\overline{0} = 4\mathbb{Z} + 0 = \{ \dots, -12, -8, -4, 0, 4, 8, 12, \dots \}$

$\overline{1} = 4\mathbb{Z} + 1 = \{ \dots, -11, -7, -3, 1, 5, 9, 13, \dots \}$

$\overline{2} = 4\mathbb{Z} + 2 = \dots$

$\overline{3} = 4\mathbb{Z} + 3 = \dots$

$$-3 = 4(-1) + 1$$

Some important properties of cosets

Let $a \in G$ and H a subgroup of G . Then:

1. $H1 = H = 1H$, that is, H is a right and left coset of itself.
2. It is obvious that $a \in Ha$ and $a \in aH$ since $a = 1 \cdot a = a \cdot 1$ and $1 \in H$.
3. If G is abelian, then $Ha = aH$ for all $a \in G$ and all subgroups H of G .
For any $h \in H$, $ha = ah$
4. 3 might not be true if G is not abelian.

Take $G = S_3$ and $H = \{e, (1\ 3)\}$, $a = (1\ 2)$

$$(1\ 2)H \neq H(1\ 2).$$

5. If $a \in Hb$, then $Ha = Hb$. ($b \in G$)

Since Ha and Hb are sets, we need to show that

$$Ha \subseteq Hb \text{ and } Hb \subseteq Ha.$$

From $a \in Hb$, we get that $a = hb$ for some element $h \in H$. Take $x \in Ha$, say $x = h_1 a$, for some $h_1 \in H$, and let us show that $x \in Hb$.

$$\underline{x} = h_1 a = h_1 (h b) = \underbrace{(h_1 h)}_e b \in \underline{Hb} \checkmark$$

To prove that $\underline{Hb} \stackrel{?}{=} \underline{Ha}$, use that
 $\boxed{b = h^{-1}a}$

→ **6.** $\underline{Ha = Hb}$ if and only if $ab^{-1} \in H$.

7. $Ha = H$ if and only if $a \in H$. (Take $b = 1$)

In additive notation **6** and **7** become

$$H + a = H + b \Leftrightarrow a - b \in H; \quad H + a = H \Leftrightarrow a \in H$$

For instance, $6\mathbb{Z} + 7 = 6\mathbb{Z} + 1$ since $7 - 1 = 6 \in 6\mathbb{Z}$.

Proof of ⑥

$\Leftarrow$ | Assume that $ab^{-1} \in H$. Then $ab^{-1} = h$ for some $h \in H$. From here we obtain that $a = hb \in Hb$. Applying ⑤ we get that $\underline{Ha = Hb}$.

$\Rightarrow$ | Suppose now that $\underline{Ha = Hb}$. Then $a = 1 \cdot a$, $1 \in H$
 $\underline{a} \in \underline{Ha} = \underline{Hb}$, and we can find $h \in H$
 such that $\underline{a = hb}$. But then $\underline{ab^{-1} = h} \in H$,
 as required.

8. There exists a bijective map from H to Ha . In particular, if H is finite, then any coset Ha has the same number of elements as H and therefore all the cosets have the same number of elements.

Define $\varphi: H \longrightarrow Ha$
$$h \longmapsto \varphi(h) = ha$$

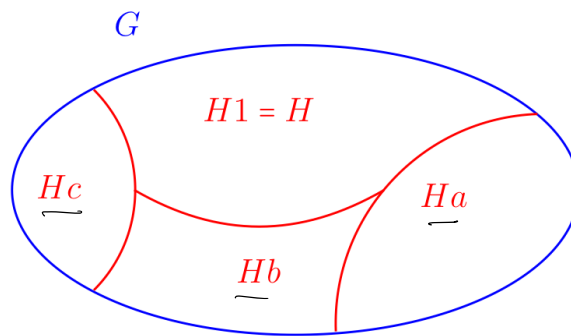
This map is surjective because any element of Ha is of the form $\underline{h_1}a$, $h_1 \in H$, and $\varphi(h_1) = h_1a$.

φ is injective: $\varphi(h) = \varphi(h') \stackrel{?}{\Rightarrow} h = h'$

$\varphi(h) = \varphi(h') \Rightarrow \underline{ha = h'a} \Rightarrow \underline{h = h'}$ ✓

↑
Cancellation
law in G
or multiplying
by $\underline{a^{-1}}$

9. The family of all the cosets of H , as a ranges over G , is a partition of G . It means that $G = \cup_{a \in G} Ha$, and $Ha \neq Hb$ implies $Ha \cap Hb = \emptyset$.



Notice that G is the union of all the cosets since for any $g \in G$, we have $g \in Hg$ (because $g = 1 \cdot g$).